- Recall that the determinant of a 2×2 matrix, $A=[a_{ij}]$, is the number $\det A = a_{11}a_{22} - a_{12}a_{21}$
- For a 1×1 matrix – say, $A=[a_{11}]$ – we define $\det A = a_{11}$.
- To generalize the definition of the determinant to larger matrices, we will use 2×2 determinants to rewrite the 3×3 determinant Δ described above.
- Since the terms in Δ can be grouped as $(a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32}) - (a_{12}a_{21}a_{33} - a_{12}a_{23}a_{31}) + (a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31})$,

$$\Delta = a_{11}\cdot\det\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33}\end{bmatrix} - a_{12}\cdot\det\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33}\end{bmatrix} + a_{13}\cdot\det\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32}\end{bmatrix}$$

For brevity, write $\boxed{\Delta = a_{11}\cdot\det A_{11} - a_{12}\cdot\det A_{12} + a_{13}\cdot\det A_{13}}$ (3)

where A_{11}, A_{12} and A_{13} are obtained from A by deleting the first row and one of the three columns.

- For any square matrix A, let A_{ij} denote the submatrix formed by deleting the ith row and jth column of A. For instance, if

$$A = \begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0\end{bmatrix}$$ then A_{32} is obtained by crossing out row 3 and column 2, $\begin{bmatrix} 1 & \cancel{-2} & 5 & 0 \\ 2 & 0 & 4 & -1 \\ \cancel{3} & \cancel{1} & \cancel{0} & \cancel{7} \\ 0 & 4 & -2 & 0\end{bmatrix}$

so that $A_{32} = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0\end{bmatrix}$.

In general, an $n\times n$ determinant is defined by determinants of $(n-1)\times(n-1)$ submatrices.

Definition: For $n\geq 2$, the **determinant** of an $n\times n$ matrix $A=[a_{ij}]$ is the sum of n terms of the form $\pm a_{ij}\det A_{ij}$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \ldots, a_{1n}$ are from the first row of A. In symbols,

$$\det A = a_{11}\det A_{11} - a_{12}\det A_{12} + \cdots + (-1)^{1+n} a_{1n}\det A_{1n}.$$

$$= \sum_{j=1}^{n} (-1)^{1+j} a_{1j}\det A_{1j}$$